

qALU: efficient error protection in quantum arithmetic circuits

Zirui Li

09/30/2026

Outline

1. Classical Reed–Muller code
2. Quantum Reed–Muller code
3. Eastin–Knill theorem
4. Universality through code switching

Outline

- 1. Classical Reed–Muller code**
2. Quantum Reed–Muller code
3. Eastin–Knill theorem
4. Universality through code switching

Repetition code: $\text{RM}(0, 3) = [8, 1, 8]$

8 physical bits, 1 message bit, distance 8

Two equivalent parity-check matrices

$$\underbrace{\begin{matrix} 1 \\ z_0 \\ z_1 \\ z_2 \\ z_0 z_1 \\ z_0 z_2 \\ z_1 z_2 \end{matrix} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}}_{H_{\text{RM}}} \longleftrightarrow_{\text{equivalent}} \underbrace{\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}}_H$$

$z_i z_j$: bitwise AND of two variable rows. $Hv^T = 0 \pmod{2}$.

Encoding uses a different matrix

$$G = (1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1), \quad v = mG$$

$$0 \mapsto 00000000 \quad 1 \mapsto 11111111$$

Single parity bit code: $RM(2, 3) = [8, 7, 2]$

8 physical bits, 7 message bits, distance 2

Append one parity bit

$$(m_0, \dots, m_6) \mapsto (m_0, \dots, m_6, p), \quad p = m_0 \oplus \dots \oplus m_6$$

Generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

Parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$Hv^T = 0 \pmod{2}$$

$$1000000 \mapsto 10000001$$

$$1100000 \mapsto 11000000$$

A single parity check detects any single-bit error.

Another classical code: $\text{RM}(1, 3) = [8, 4, 4]$

8 physical bits, 4 message bits, distance 4

Generator matrix

$$G = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$v = mG \pmod{2}$$

$$Hv^T = 0 \pmod{2}$$

Self-dual code: we can choose $G = H$.

Message	$\mapsto$	Codeword
1000	$\mapsto$	11111111
0100	$\mapsto$	00001111
0010	$\mapsto$	00110011
0001	$\mapsto$	01010101

Classical encoding: four message bits become eight bits

Message	$\mapsto$	Codeword
0000	$\mapsto$	00000000
1000	$\mapsto$	11111111
0100	$\mapsto$	00001111
0010	$\mapsto$	00110011
0001	$\mapsto$	01010101

Linear code

$$\begin{array}{rcl} 0100 & \mapsto & 00001111 \\ + & & + \\ 0010 & \mapsto & 00110011 \\ \hline = 0110 & \mapsto & 00111100 \end{array}$$

Three basis vectors span a subspace

Message		Codeword
0000	⟶	00000000
1000	⟶	11111111
0100	⟶	00001111
0010	⟶	00110011
0001	⟶	01010101

These three basis vectors span a subspace.

Use three message bits:

Message	⟶	Codeword
000	⟶	00000000
100	⟶	00001111
010	⟶	00110011
001	⟶	01010101

Classical automorphism: swap logical bits 1 and 2

$$(m_1, m_2, m_3) \longleftrightarrow (m_2, m_1, m_3)$$

Swap physical bit positions: $q_2 \leftrightarrow q_4$ $q_3 \leftrightarrow q_5$

Message	Codeword							
000 $\mapsto$	0	0	0	0	0	0	0	0
100 $\mapsto$	0	0	0	0	1	1	1	1
010 $\mapsto$	0	0	1	1	0	0	1	1
001 $\mapsto$	0	1	0	1	0	1	0	1
Before	q_0	q_1	q_2	q_3	q_4	q_5	q_6	q_7
After	q_0	q_1	q_2	q_3	q_4	q_5	q_6	q_7



Logical: $100 \longleftrightarrow 010$ Physical: $00001111 \longleftrightarrow 00110011$

Classical automorphism: add logical bit 1 to bit 2

$$(m_1, m_2, m_3) \longleftrightarrow (m_1, m_2 \oplus m_1, m_3)$$

Swap physical bit positions: $q_2 \leftrightarrow q_6$ $q_3 \leftrightarrow q_7$

Message	Codeword							
000 $\mapsto$	0	0	0	0	0	0	0	0
100 $\mapsto$	0	0	0	0	1	1	1	1
010 $\mapsto$	0	0	1	1	0	0	1	1
001 $\mapsto$	0	1	0	1	0	1	0	1
Before	q_0	q_1	q_2	q_3	q_4	q_5	q_6	q_7
After	q_0	q_1	q_2	q_3	q_4	q_5	q_6	q_7

Logical: $100 \longleftrightarrow 110$

Physical: $00001111 \longleftrightarrow 00111100$

Outline

1. Classical Reed–Muller code
- 2. Quantum Reed–Muller code**
3. Eastin–Knill theorem
4. Universality through code switching

From classical rows to quantum stabilizer checks

Valid encoded states satisfy

$$S|\psi_L\rangle = |\psi_L\rangle.$$

Checks detect errors without reading the logical data.

All-X XXXXXXXX

$Z(1)$ ZZZZZZZZ

$Z(z_0)$ IIIIZZZZ

$Z(z_1)$ IIZZIIZZ

$Z(z_2)$ IZIZIZIZ

8 physical qubits – 5 independent checks = 3 logical qubits

$[[8, 3, 2]]_Z$: detects any single-qubit Pauli error.

X flips bits; Z flips phase

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle$$

Bit flips: $XXXXXXX = X^{\otimes 8}$

$$XXXXXXX |01100100\rangle = |10011011\rangle \quad \text{not stabilized}$$

$$XXXXXXX \frac{|01100100\rangle + |10011011\rangle}{\sqrt{2}} = \frac{|10011011\rangle + |01100100\rangle}{\sqrt{2}} \quad \text{stabilized}$$

Phase flips: $IIIZZ = I^{\otimes 4} \otimes Z^{\otimes 4}$

$$IIIZZ |10000001\rangle = -|10000001\rangle \quad \text{not stabilized}$$

$$IIIZZ |10000011\rangle = +|10000011\rangle \quad \text{stabilized}$$

Stabilized by S means $S|\psi\rangle = +|\psi\rangle$.

Three logical bits are polynomial coefficients

Polynomial	Physical bits
z_0	00001111
z_1	00110011
z_2	01010101

$$v_b = b_0 z_0 + b_1 z_1 + b_2 z_2$$

$$\overline{|b_0 b_1 b_2\rangle}_L = \frac{|v_b\rangle + |v_b \oplus 11111111\rangle}{\sqrt{2}}$$

$$\overline{|000\rangle}_L = (|00000000\rangle + |11111111\rangle)/\sqrt{2},$$

$$\overline{|100\rangle}_L = (|00001111\rangle + |11110000\rangle)/\sqrt{2},$$

$$\overline{|010\rangle}_L = (|00110011\rangle + |11001100\rangle)/\sqrt{2},$$

$$\overline{|001\rangle}_L = (|01010101\rangle + |10101010\rangle)/\sqrt{2},$$

$$\overline{|110\rangle}_L = (|00111100\rangle + |11000011\rangle)/\sqrt{2}.$$

Quantum automorphism: logical SWAP

$$\overline{|b_0 b_1 b_2\rangle}_L \longleftrightarrow \overline{|b_1 b_0 b_2\rangle}_L$$

Swap physical qubits

$$q_2 \leftrightarrow q_4 \quad q_3 \leftrightarrow q_5$$

$$\overline{|100\rangle}_L \longleftrightarrow \overline{|010\rangle}_L$$



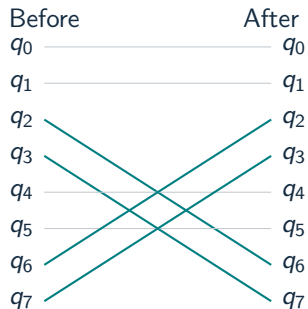
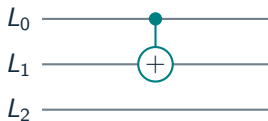
Quantum automorphism: logical CNOT

$$\overline{|b_0 b_1 b_2\rangle}_L \mapsto \overline{|b_0, b_1 \oplus b_0, b_2\rangle}_L$$

Swap physical qubits

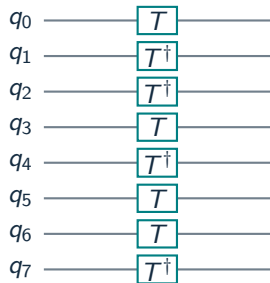
$$q_2 \leftrightarrow q_6 \quad q_3 \leftrightarrow q_7$$

$$\overline{|100\rangle}_L \mapsto \overline{|110\rangle}_L$$

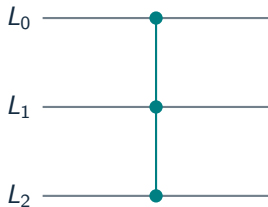


The $[[8, 3, 2]]$ cube code also has a native logical CCZ

Physical phase layer



Logical CCZ



$$|b_0 b_1 b_2\rangle_L \mapsto (-1)^{b_0 b_1 b_2} |b_0 b_1 b_2\rangle_L$$

Four physical T gates and four $T^\dagger$ gates.

Outline

1. Classical Reed–Muller code
2. Quantum Reed–Muller code
- 3. Eastin–Knill theorem**
4. Universality through code switching

Universal

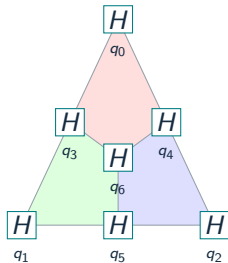
- Clifford gates: H , S , CNOT, $\sqrt{X}$, $\sqrt{Y}$, CZ, ...
- Non-Clifford gates: T , CCX, CS, ...

Clifford gates + one non-Clifford gate type
(Clifford + T , Clifford + CCX, Clifford + CS, ...)
gives universal quantum computing.

What is transversal gates?

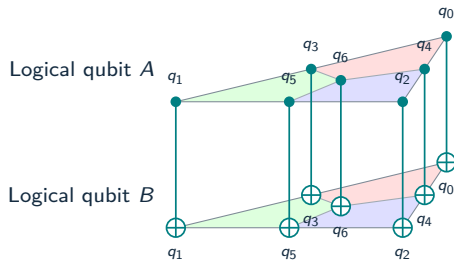
Take $[[7, 1, 3]]$ color code for example.

Hadamard: one gate per physical qubit



$$\overline{H} = H^{\otimes 7}$$

CNOT: pair corresponding qubits



$$\overline{\text{CNOT}}_{A \rightarrow B} = \prod_{i=0}^6 \text{CNOT}_{A_i \rightarrow B_i}$$

Each physical gate touches at most one qubit in each code block.

IBM Quantum Learning: controlling error propagation.

Eastin–Knill: no universal transversal gate set

Eastin–Knill theorem

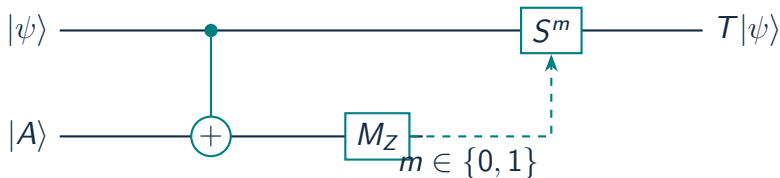
A fixed code that detects arbitrary single-qubit errors cannot have a universal set of transversal logical gates.

Code	Clifford gates	T gate: standard route
$[[7, 1, 3]]$ color code	Transversal H , S , CNOT	Magic-state injection
2D surface code	Code deformation / lattice surgery	Magic-state injection

$[[8, 3, 2]]$ can do transversal CCZ and free CNOTs,
but cannot do transversal H and S .

Magic state

$$|A\rangle = T|+\rangle = \frac{|0\rangle + e^{i\pi/4}|1\rangle}{\sqrt{2}}$$



A standard route to universality: inject a non-Clifford magic state using Clifford gates and measurement.

Magic-state injection; Bravyi and Kitaev, arXiv:quant-ph/0403025.

Outline

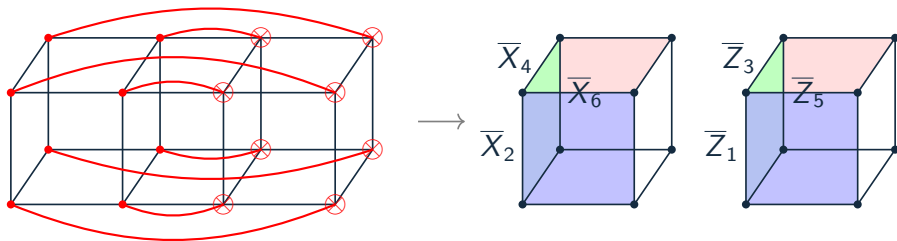
1. Classical Reed–Muller code
2. Quantum Reed–Muller code
3. Eastin–Knill theorem
- 4. Universality through code switching**

A universal gate set through code switching

$$\underbrace{\{H, S, \text{CNOT}\}}_{\text{Clifford}} + \text{CCZ} \implies \text{universal quantum computing}$$

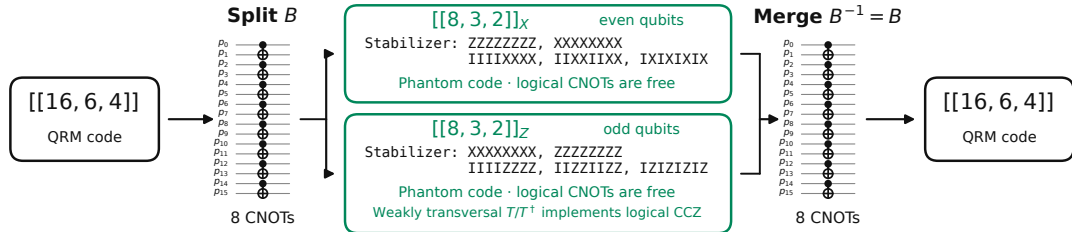
CNOT by permutation; CCZ by a physical $T/T^\dagger$ layer.

How do we implement H and S ?



Code switching between $[[16, 6, 4]]$ and $[[8, 3, 2]]$.

Split one tesseract into two phantom factors



$$B = \prod_{y=0}^7 \text{CNOT}_{2y \rightarrow 2y+1}, \quad B^{-1} = B.$$

Eight CNOTs to split; eight to merge. The factors each have distance two.

From qrmfold to Monte Carlo tree search

- **qrmfold** constructs logical H and S on the $[[16, 6, 4]]$ tesseract code using transversal and fold-transversal layers.
- **316 involutive automorphisms** supply two fold types:

$$316 \times 2 + 1 = \mathbf{633} \text{ cheap logical operations}$$

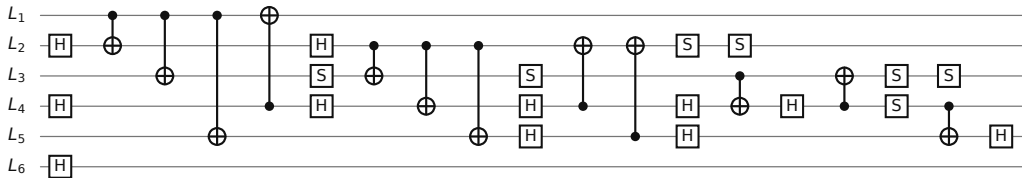
- Each cheap logical operation can implement several $H/S/CZ/CNOT$ gates together.

We use the 633 available cheap logical operations as a large gate set and propose **Monte Carlo tree search (MCTS)** to reduce compilation overhead.

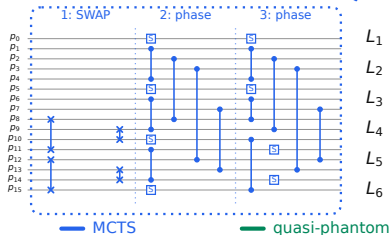
Find a cheap sequence of logical operations + a cheaper remaining Clifford.

A worked target drops from 3,368 to 372 two-qubit gates

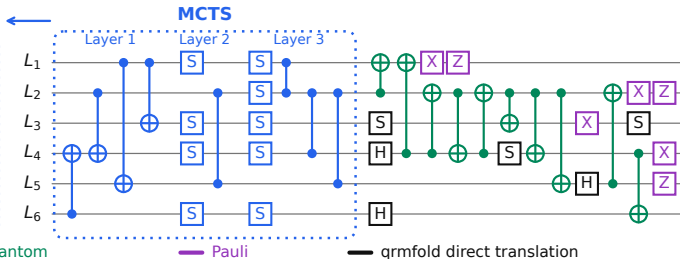
(a) Target Clifford



(c) Physical MCTS layers



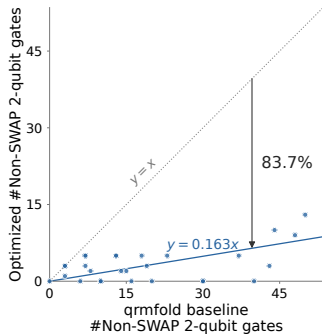
(b) MCTS + quasi-phantom



Reducing physical two-qubit gate counts

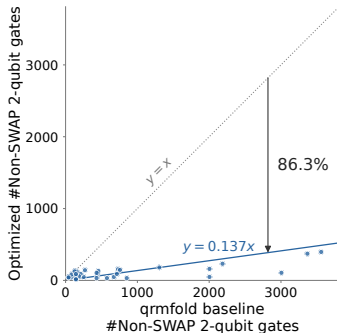
[[4, 2, 2]]

MCTS



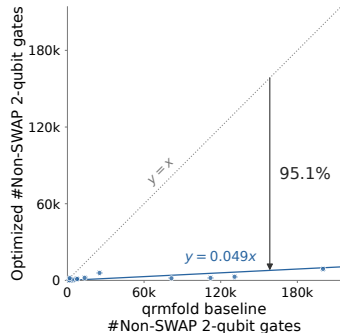
[[16, 6, 4]]

MCTS + quasi-phantom



[[64, 20, 8]]

Quasi-phantom



Below the diagonal: fewer gates than direct qrmfold.

Physical non-SWAP two-qubit gates; permutations tracked in the wire map.

Evaluate accepted results, not only gate counts

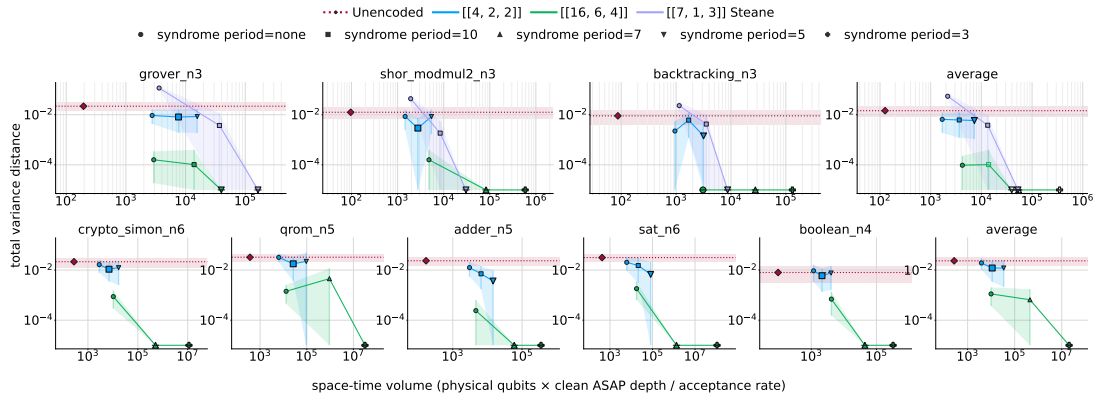
$$\text{TVD} = \frac{1}{2} \sum_x |P(x) - Q_{\text{acc}}(x)|$$

$$\text{Space-time volume} = \frac{N_{\text{phys}} D}{a}, \quad a = \text{Pr}(\text{accept})$$

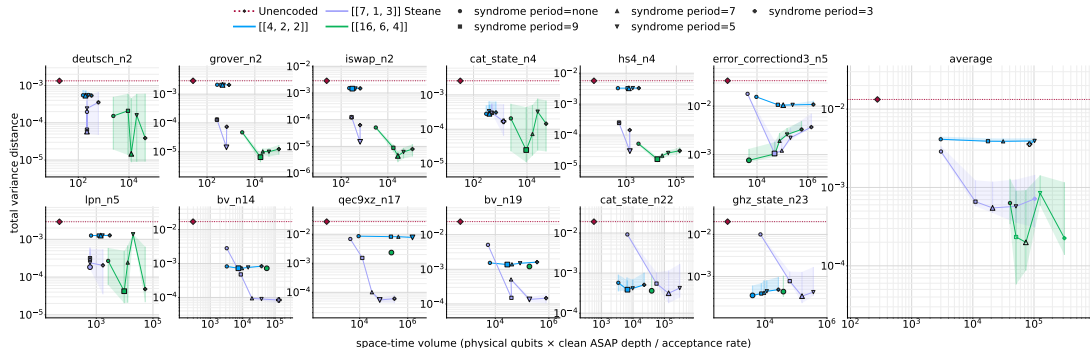
- 12 Clifford workloads; 8 Clifford+CCZ kernels. No full Shor run.
- Gate noise: $p_2 = 10^{-3}$, $p_1 = 10^{-4}$; preparation and checks included.

Lower output error can require more rejected attempts.

Clifford + CCZ: error versus space-time volume



Clifford: error versus space-time volume



Q&A